

Chapter 1 — Introduction

1.1 Background and Motivation

Across most of human evolutionary history, socioeconomic resources and reproductive success co-varied positively: individuals with greater access to land, livestock, or social standing produced more surviving offspring than those with less. This pattern, predicted by the core models of human behavioural ecology, is well documented across pre-industrial agrarian and forager populations (Boone, 1986; Volland, 1990; Hopcroft, 2006; Nettle and Pollet, 2008). Beginning in the eighteenth and nineteenth centuries, however, the demographic transition decoupled these variables across most of Europe and subsequently the rest of the world. In contemporary high-income societies the relationship has often inverted: those with the highest education and income have, on average, the fewest children. Vining (1986) named this the central theoretical problem of human sociobiology, and despite four decades of subsequent research the underlying mechanisms remain contested.

The puzzle is not merely descriptive but theoretical. No equivalent reversal has been documented in any other species, and the human pattern therefore requires a specifically human explanation. Three families of explanation compete in the contemporary literature, each making distinct predictions about the conditions under which the gradient should appear and the populations among which it should be steepest. The first is the cultural transmission framework, which argues that low-fertility norms diffuse downwards through social networks from higher-status individuals to lower-status individuals (Bongaarts and Watkins, 1996; Colleran, 2016). The second is the quantity–quality framework, which argues that modern returns to human capital make concentrating investment on fewer children the fitness-maximising strategy (Kaplan, 1996; Lawson and Mace, 2011). The third is the maladaptation framework, which treats modern low fertility as a mismatch between evolved psychological mechanisms and environments with effective contraception, extended education, and welfare

provision (Pérusse, 1993; Kanazawa, 2003). These frameworks are not strictly mutually exclusive, but they make different empirical predictions and have been invoked to support different theoretical conclusions about the future of human fertility.

A more recent literature, beginning with Hopcroft (2006, 2015) and consolidated by Stulp and Barrett (2016), has questioned whether the socioeconomic-fertility relationship is in fact uniformly negative. Several studies have documented that fertility ceases to decline, or rises again, at the upper end of the SES distribution; a so-called J-curve. Hopcroft, using US General Social Survey data, found positive fertility–income gradients among married high-income men once unmarried childlessness was accounted for. Fieder and Huber (2007) found analogous patterns in Swedish men. Nettle and Pollet (2008) reported similar results across several Western populations. If this revival is genuine, the paradox may be more limited in scope than originally framed, affecting the middle of the distribution but not its extremes. This would imply that the psychology of resource-to-fertility conversion is partially intact, and that low fertility among middle-SES populations reflects specific features of their economic and cultural environment rather than a general maladaptation.

Whether the J-curve is genuine, where it appears first, and what mechanisms drive it remain open questions of direct relevance both to evolutionary theory and to demographic forecasting. The existing evidence, however, is methodologically uneven. Few studies use education-stratified fertility data across multiple countries, and modern distributional methods such as quantile regression and semi-parametric splines are rarely deployed. Most published evidence relies on cross-sectional surveys, single-country designs, or period-based measures, leaving the cross-national robustness of the J-curve substantially unsettled.

1.2 Research Problem

The contemporary literature on the socioeconomic-fertility gradient suffers from three interlocking gaps. First, despite the theoretical importance of the J-curve hypothesis, systematic tests using education-stratified fertility data across multiple European countries are lacking. Existing evidence is fragmentary, drawn from disparate countries with disparate measures, and rarely uses formal distributional methods to characterise the full shape of the gradient. Second, cross-country comparisons of cultural and economic mechanisms typically rely on a small number of pre-selected covariates in low-sample country-level regressions, an approach vulnerable to both omitted-variable bias and overfitting. Third, individual-level mediation analyses of the education effect on fertility, particularly those that decompose the effect into attitudinal and economic pathways, have been conducted only in fragmentary form, and rarely with modern methods for handling high-dimensional pre-treatment confounding. The British cohort studies, and the 1970 British Cohort Study (BCS70) in particular, have been used extensively in adjacent literatures but remain substantially underused for this specific question, despite the Age 51 sweep now providing essentially complete fertility outcomes for the 1970 birth cohort.

The present dissertation addresses these gaps through a four-part empirical strategy operating at descriptive, distributional, country-level, and individual-level scales. The central research question is:

What is the shape of the contemporary socioeconomic-fertility gradient across European populations, and what conditions predict its steepness and the location of any non-monotonicities?

1.3 Research Questions and Hypotheses

The central question decomposes into four testable hypotheses. Together, they span descriptive pattern, distributional shape, country-level mechanism, and individual-level mechanism.

H1 (Inversion). Across European populations observed between 2007 and 2024, the relationship between education and completed fertility is negative; that is, women with higher educational attainment have lower completed fertility than women with lower educational attainment, after controlling for country- and year-specific factors.

H2 (Non-monotonicity). The education–fertility relationship is non-monotonic across the distribution of educational attainment, with fertility ceasing to decline, or rising again, at the upper end of the distribution.

H3 (Cultural transmission). Cross-country variation in the steepness and shape of the gradient is predicted better by cultural indicators: secularisation, individualism, gender-norm change, than by economic indicators such as GDP per capita, female labour-force participation, and contraceptive prevalence.

H4 (Individual mechanism). Within individuals in the 1970 British Cohort Study, the effect of education on completed fertility is mediated primarily by attitudinal and preference variables rather than by realised earnings.

The hypotheses are designed to discriminate between the three principal theoretical frameworks rather than to confirm any one of them. The dissertation does not aim to resolve the fitness paradox theoretically; it aims to produce sharp empirical results that constrain which existing explanations remain viable. Specifically, H1 and H2 test the descriptive claim that motivates the paradox itself; H3 tests the cultural transmission account at the country level; H4 tests it at the individual level. A pattern of results in which the gradient is non-monotonic (H2 supported), in which cultural indicators dominate cross-country variation (H3 supported), and in which attitudinal pathways dominate individual-level mediation (H4 supported) would

constitute strong evidence for the cultural transmission framework. The opposite pattern would constitute strong evidence for the quantity–quality framework. Intermediate patterns; which the analyses in fact recover — require more nuanced interpretation, addressed in detail in Chapter 7.

1.4 Theoretical Framework

The dissertation engages three theoretical traditions without committing in advance to any one. From classical demographic transition theory it adopts the period–cohort distinction and the formal apparatus of education-stratified fertility (Notestein, 1945; Caldwell, 1976). From human behavioural ecology it adopts the predictive framework of resource–fertility correlations and the quantity–quality tradeoff (Kaplan, 1996; Borgerhoff Mulder, 1998). From cultural evolutionary theory it adopts the model of fertility norms as socially transmitted traits subject to prestige and conformity biases (Colleran, 2016; Mace, 2008). The empirical strategy is designed to discriminate between these frameworks rather than to assume any one of them. The integration is methodological as well as substantive: classical fixed-effects panel methods, modern distributional methods, interpretable machine learning, Bayesian inference, and double machine learning for causal mediation are each matched to a hypothesis and to a scale of analysis. The substantive contribution depends on the methodological one; modern computational tools are deployed in service of an old theoretical question rather than as ends in themselves.

1.5 Contribution

The dissertation makes three substantive contributions to the literature on the demographic transition fitness paradox. First, it provides a methodologically modern multi-country test of the non-monotonicity hypothesis. Using education-stratified period fertility data across twenty-one European countries between 2007 and 2024, the analysis tests for U-shaped or J-shaped gradients using both parametric (quadratic fixed-effects) and semi-parametric (quantile regression) methods. This goes beyond the single-country, single-method designs that have characterised the existing J-curve literature and produces a defensible cross-national characterisation of the shape of the gradient.

Second, the dissertation provides a principled high-dimensional comparison of cultural and economic explanations of cross-country variation in the gradient. Combining gradient-boosted variable-importance methods (XGBoost with SHAP values) with Bayesian hierarchical regression - an exploratory-then-confirmatory pipeline that separates discovery from inference, the analysis sidesteps both the small-N regression problems and the discretionary covariate-selection problems that have constrained existing cross-country work. To my knowledge, this represents the first systematic high-dimensional comparison of cultural and economic predictors of fertility variation in the European context.

Third, the dissertation provides an individual-level mediation analysis of the education–fertility relationship using double machine learning (Chernozhukov et al., 2018) on the BCS70 cohort. To my knowledge, no published analysis has formally decomposed the education effect into attitudinal, economic, and partner-related pathways using modern methods for high-dimensional confounding adjustment on this cohort. The recently released Age 51 sweep provides essentially complete fertility outcomes for the first time, making this an opportune moment for such an analysis.

A secondary contribution is methodological. The dissertation demonstrates the responsible integration of interpretable machine learning with classical demographic methods and causal inference in a substantively-driven empirical project. Each computational tool is matched to a specific role: panel econometrics for descriptive identification, quantile and quadratic methods for distributional shape, gradient boosting for flexible feature discovery, SHAP values for interpretation, Bayesian hierarchical models for uncertainty quantification, and double machine learning for high-dimensional confounding adjustment in mediation. The aim is to demonstrate that modern computational methods can be deployed in service of substantive social-science questions without displacing them.

1.6 Findings in Brief

The four hypotheses receive heterogeneous support, and the resulting picture is more complex than the simple negative-gradient framing of much existing literature. The descriptive analysis reveals that, across the twenty-one countries studied, the relationship between education and period fertility is U-shaped rather than monotonic. Medium-educated women exhibit the lowest fertility in most countries, with both low- and high-educated women showing higher fertility. The U-shape is statistically robust to alternative specifications, holds across the conditional fertility distribution under quantile regression, and is significantly stronger than a linear gradient on standard model-fit criteria. The high-education contrast against low education is negative on average, partially supporting H1, but the strict monotonic prediction is rejected in the great majority of countries. H2 is therefore confirmed in a stronger form than originally specified: the relationship is non-monotonic, but as a U-shape across the full distribution rather than only a J-curve at its upper tail.

The cross-country analysis under H3 produces a mixed verdict. The variable-importance results from XGBoost with SHAP slightly favour cultural over economic predictors, but the Bayesian hierarchical analysis finds the strongest individual signal in GDP per capita, an

economic indicator. Neither family of explanations clearly dominates at the country level, and substantial uncertainty remains given the small effective sample of twenty-one countries.

The individual-level mediation analysis under H4 produces the most theoretically surprising result. Contrary to the hypothesis derived from Snopkowski and Sear (2013) and the broader cultural-transmission literature, the economic pathway - opportunity costs operating through realised earnings, accounts for the majority of the education effect on fertility, while the attitudinal pathway contributes comparatively little. This finding supports the quantity-quality framework over the cultural transmission framework at the individual level, at least for the British 1970 cohort, and runs against the prediction of the dissertation's own H4. Taken together, these findings support a more nuanced reading of the fitness paradox than the existing literature typically offers, with implications discussed at length in Chapter 7.

1.7 Dissertation Structure

The remainder of the dissertation proceeds as follows. Chapter 2 reviews the literature on the demographic transition, the SES-fertility gradient, the three principal theoretical frameworks, the J-curve hypothesis, and individual-level mechanism studies, situating the four hypotheses against the existing evidence and identifying the methodological and substantive gaps the present work addresses. Chapter 3 details the data and methodology: the five datasets used (Eurostat, the Wittgenstein Centre, V-Dem, the OECD and World Bank, and BCS70), the construction of analytical variables, the methods deployed for each hypothesis, and the methodological decisions made during data preparation. Chapter 4 presents the descriptive results: cross-country and within-country patterns of fertility by education, the shape of the gradient by country, and the heterogeneity that motivates the formal analyses. Chapter 5 presents the main results for H1 and H2 — two-way fixed-effects panel models, quadratic specifications, and quantile regression across the fertility distribution. Chapter 6 presents the mechanism analyses for H3 and H4 — XGBoost with SHAP values, Bayesian hierarchical

models with regularising priors at the country level, and double machine learning causal mediation at the individual level. Chapter 7 discusses the substantive and theoretical implications of the findings, their relation to the existing literature, the strengths and limitations of the analysis, and directions for future research. Chapter 8 concludes by restating the research question, summarising the findings, and stating the contribution to the literature on the demographic transition fitness paradox.